

HITESH SHIMPI

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SUMMARY

Computer Science student skilled in Backend Development, Machine Learning, and Deep Learning, with hands-on experience building scalable APIs and AI-driven applications. Experienced in Python, FastAPI, REST APIs, AWS, Data Engineering, with a strong interest in Backend Systems, Docker-based Containerization, and Cloud Infrastructure.

EDUCATION

- **SRM Institute of Science and Technology** Aug 2023 - May 2027
Bachelor of Technology in Computer Science and Engineering Chennai, India
- **KKJR Vidyalaya** Jun 2022 - May 2023
Class XII Chalisgaon, India
- **St. Joseph Convent School** Apr 2020 - Apr 2021
Class X Chalisgaon, India

EXPERIENCE

AICTE | AI/ML Developer Intern

April 2024 – June 2024

- Developed a deep learning-based **rice leaf disease detection system** using EfficientNetB0 for 6-class image classification, with 224×224 image preprocessing, normalization, and data augmentation.
- Enhanced feature representation by integrating **CBAM** attention and **GeM pooling**, improving the model's ability to capture discriminative features from rice leaf images.
- **Tech Stack:** Python, TensorFlow, Keras, EfficientNetB0, CBAM, GeM Pooling, Adam Optimizer

PROJECTS

MRD Classification [↗](#)

Tech Stack: Python, TensorFlow, Keras, EfficientNet, DenseNet, Transfer Learning, Grad-CAM

- Engineered a deep learning-based **multi-label retinal disease classification** system using EfficientNet and DenseNet on **5,000+ retinal images** to classify multiple disease categories.
- Applied **transfer learning** with EfficientNet and DenseNet to develop the classification models, leveraging pretrained feature representations for multi-label disease classification.
- Applied **Grad-CAM** visualizations to interpret model predictions, highlight disease-relevant regions, and provide visual insights into the model's decision-making across multiple retinal disease classes.
- Integrated model interpretability techniques to provide visual insights into classification decisions, analyze disease-relevant features, and improve the transparency and explainability of deep learning predictions.

Edulink [↗](#)

Tech Stack: React.js, Tailwind CSS, Node.js, Express.js, FastAPI (Python), REST APIs, SQL

- Built a full-stack collaborative study platform supporting real-time messaging, study groups, resource sharing, and user interactions.
- Implemented backend services and REST APIs using FastAPI and integrated frontend components with backend services for seamless user interactions.
- Optimized database queries and backend operations, reducing application response latency by 30% maintaining efficient data access.
- Designed a scalable application architecture integrating React.js frontend, FastAPI backend, and SQL-based data management for collaborative learning workflows.

SKILLS

- **Languages:** JavaScript, Python, SQL
- **Frameworks:** React.js, FastAPI, Node.js, Tailwind CSS, Next.js, Vue.js.
- **Databases:** MySQL, PostgreSQL
- **Tools & Technologies:** Git, REST API, Docker, Postman, Supabase, Apache Spark, PySpark, ETL.
- **Others:** AWS, TensorFlow, Keras, PyTorch, Scikit-learn, CNNs, Databricks Lakeflow
- **Soft Skills:** Problem Solving, Team Collaboration, Strong Communication

ACHIEVEMENTS & CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner
- Databricks Badges: Apache Spark, Lakeflow, Data Pipelines, ETL
- **Patent:** Smart Blind Spot Detection for Vehicles Using Multi-Sensor Integration and AI-Based Warning Mechanisms

HACKATHONS

Guidewire DEVTRAILS '26. • Accenture Databricks '26. • Infosys HackWithInfy '26. • Capgemini Brand Quest

EXTRA CURRICULAR

- Core team member of Team ENVISION (AARUUSH SRMIST) - organized ML/AI workshops and tech sessions, fostering data-driven learning culture on campus.
- Active programmer with expertise in Python and Backend with consistent algorithmic problem-solving across LeetCode, and HackerRank.
- Exploring video editing and content creation while continuously learning and developing creative post-production skills.